
When Does Shift-Invariance Permit Adversarial Robustness?

A Margin-Preservation Theory

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Abstract

Making a convolutional network more shift-invariant has been reported both to reduce its adversarial robustness [10] and to improve it [12, 31]. We show that these are two regimes of one quantity: the margin of the discriminative signal that is preserved when the data is projected onto the invariant feature space, measured against that feature’s sensitivity, written as a margin-to-Lipschitz ratio η/L . When the projection keeps a large ratio, invariance and robustness coexist; when it leaves only a small margin, they trade off. For a shift-invariant linear classifier the ratio is exact and recovers the $1/\sqrt{d}$ robustness collapse of Ge et al. [10] for a localized signal, while for frequency-coded classes a convolution-plus-pooling layer with a nonlinearity computes power-spectrum features that separate them when no linear classifier can. The ratio is also a robustness certificate, $r_2 \geq \eta/L$, which we complete to a two-sided bracket under an explicit reachability condition, and it places the apparent counterexample of Kamath et al. [18] as a small-margin corner rather than a contradiction. On the optimization side, shift-invariance cannot raise the margin on orbit-closed data, so any robustness gain there comes through the Lipschitz term, which we prove in the lazy regime. Empirically, across standard and adversarial training, three datasets, two threat models, and a PreActResNet-18 backbone, the threat-matched ratio predicts the adversarial robust radius and AutoAttack robust accuracy, whereas shift-consistency does not by itself, and under adversarial training it is negatively related to robustness. The mechanism is a decomposition of each operator’s effect into a margin term and a Lipschitz term: a graded anti-aliasing sweep improves robustness at matched clean accuracy, while exact invariance hurts by collapsing the margin, a single change that lowers clean accuracy and robustness together.

1 Introduction

Let an image classifier be *shift-invariant* if its prediction is unchanged by circular translations of the input. Ge et al. [10] showed, for a binary problem in which each class is a single image and all its shifts, that a fully shift-invariant convolutional network needs an ℓ_2 perturbation of order $1/\sqrt{d}$ to be fooled while a fully connected network needs order 1, and that this gap appears on real datasets. A separate line of architectural work points the other way. Anti-aliased pooling improves shift-consistency together with clean accuracy and corruption robustness [40], and reducing aliasing has since been reported to improve *adversarial* robustness as well [12, 28, 31]: operators that make a network more shift-invariant also make it more robust. A parallel line proves a trade-off between *spatial* (geometric) and adversarial robustness [18, 33].

These statements are not contradictory; they are different regimes of one quantity. Our contributions:

- **Structural theory (§3, Figure 1).** An exact identity for the invariant linear margin as the separation of the data projected onto the invariant subspace (Theorem 1); a co-existence/trade-off dichotomy governed by the margin-to-Lipschitz ratio η/L , which, under an explicit co-Lipschitz condition, completes to a two-sided bracket $\eta/L \leq r_2 \leq \eta/\alpha$ (a characterization up to a condition number, §3.2); a lemma that a single convolution-plus-pooling layer computes power-spectrum features and so can separate signal that linear invariance discards (Lemma 1); and a recovery of Ge et al. [10] together with an interpretation that places Kamath et al. [18] as a small- η/L corner.
- **Optimization: a margin-free theorem and the open selection problem (§4).** The structural theory establishes that robust shift-invariant classifiers exist; whether gradient descent reaches them is separate. We give the exact rank of a shift orbit (its number of nonzero Fourier coefficients), and prove that on orbit-closed data shift-invariance is *margin-free*: the invariant and unconstrained two-layer-ReLU max-margin values coincide, so weight sharing cannot raise the margin and gradient descent self-symmetrizes (the trained network becomes shift-invariant on its own). A direct synthetic test confirms this on generic orbit data (the invariant and unconstrained networks reach the same robustness), while on near-orthogonal cluster-geometry data the invariant architecture does reach a larger η/L , which we prove in the lazy/NTK regime (orbit-averaging is a non-expansive projection in the tangent RKHS), leaving the rich-regime trajectory open.
- **Experiments (§5).** A training-free test confirms the structural predictions; a controlled, capacity-matched architectural dissection on MNIST, Fashion-MNIST, and CIFAR-10 confirms that η/L , not shift-consistency, predicts the robust radius, and decomposes each invariance operator’s effect into a margin term and a Lipschitz term. Under PGD adversarial training the same two laws hold across three datasets and two threat models and scale to a PreActResNet-18 backbone, where they also hold per sample as a pointwise certificate.

Full proofs are in Appendix A; an extended optimization roadmap is in a companion technical report.

2 Preliminaries

We work with one-dimensional signals $x \in \mathbb{R}^d$; the two-dimensional case replaces $\mathbb{Z}_d$ by $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2}$ and the cyclic DFT by its two-dimensional analogue, without change to the statements. The cyclic group $G = \mathbb{Z}_d$ acts by the circular shift $(S_s x)_i = x_{(i-s) \bmod d}$; each S_s is an orthogonal permutation. The *orbit* of x is $\mathcal{O}(x) = \{S_s x\}_{s \in G}$, the set of all its shifts, and a class set is *orbit-closed* if it is closed under all shifts.

Invariant and AC subspaces. Let $V_{\text{inv}} = \{v : S_s v = v \ \forall s\}$ and $V_{\text{ac}} = V_{\text{inv}}^\perp$. The orthogonal projector onto V_{inv} is $\Pi_{\text{inv}} = \frac{1}{d} \mathbf{1}\mathbf{1}^\top$, so $V_{\text{inv}} = \text{span}(\mathbf{1})$ is the constant (DC) line and $V_{\text{ac}} = \{v : \sum_i v_i = 0\}$. Write $f_{\text{dc}}(x) = \frac{1}{\sqrt{d}} \sum_i x_i$ for the unit DC functional. All S_s are simultaneously diagonalized by the DFT; a shift multiplies Fourier coefficient k by a unit-modulus phase, so $|\hat{x}_k|^2$ is shift-invariant.

Robustness and consistency. For a score f correct on (x, y) , the ℓ_p robust radius is $r_p(f, x) = \inf\{\|\delta\|_p : \text{sign } f(x + \delta) \neq y\}$; for a linear score $w^\top x + b$ this is $y(w^\top x + b)/\|w\|_q$. The shift-consistency is $\text{SC}(f) = \Pr_{x,s}[\text{sign } f(S_s x) = \text{sign } f(x)]$; exact invariance gives $\text{SC} = 1$.

3 Structural theory: when invariance permits robustness

3.1 The exact invariant linear margin

Theorem 1 (Invariant linear margin equals projection separation). *A linear score $g(x) = w^\top x + b$ is shift-invariant for all x if and only if $w \in V_{\text{inv}}$, i.e. $w = c\mathbf{1}$. Consequently, for any finite separable classes X_+, X_- , the invariant ℓ_2 max-margin equals the separation of the DC-projected data,*

$$\gamma_{\text{inv}} = \frac{1}{2} \text{dist}(I_+, I_-) = \frac{1}{2} \text{dist}(\text{conv } \Pi_{\text{inv}} X_+, \text{conv } \Pi_{\text{inv}} X_-),$$

where $I_\pm = [\min_{x \in X_\pm} f_{\text{dc}}(x), \max_{x \in X_\pm} f_{\text{dc}}(x)]$ are the intervals of DC values, with maximizing direction $\pm \mathbf{1}/\sqrt{d}$ oriented toward the positive class. This holds for any finite separable dataset, not only the single-image case, and does not require orbit-closure.

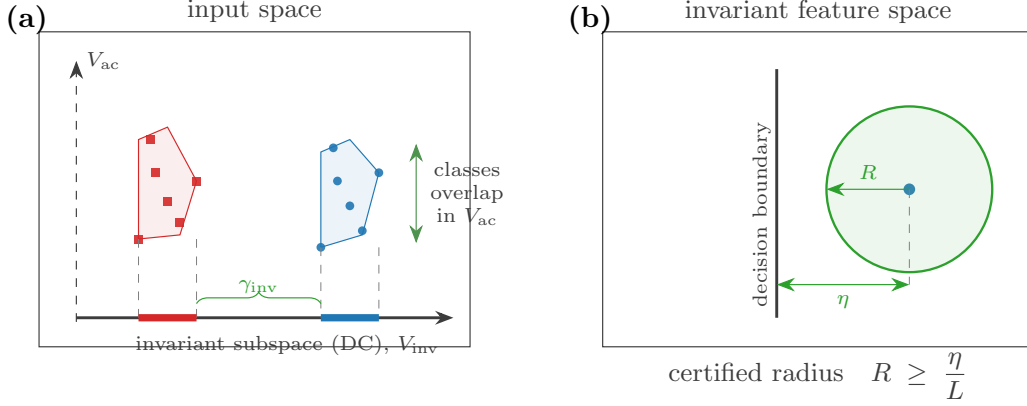


Figure 1: The central criterion. **(a)** A shift-invariant linear classifier sees the data only through its projection onto the invariant subspace V_{inv} (the DC line for cyclic shift); the separation that remains is the invariant margin γ_{inv} , the distance between the two projected class hulls (Theorem 1). Discriminative signal that lives in the orthogonal AC subspace V_{ac} is invisible to invariance, where the classes overlap. **(b)** A margin η in an invariant feature space certifies an ℓ_2 robust radius $r_2 \geq \eta/L$, with L the Lipschitz constant of the invariant feature map Ψ (§3.2); here $\eta = \gamma_{\text{inv}}$ for the linear case. Invariance helps robustness only when the discriminative signal is preserved by the projection with positive margin.

Proof sketch. Invariance forces $(S_{-s}w - w)^\top x = 0$ for all x, s , hence $w \in V_{\text{inv}} = \text{span}(\mathbf{1})$ (only constant vectors are shift-fixed). With $w = c\mathbf{1}$ the score depends on x only through $f_{\text{dc}}(x)$, reducing the problem to one-dimensional thresholding of the DC values; the max-margin separator of two point sets on a line is the midpoint of the gap between their convex hulls (here the two intervals $I_{\pm}$), so the geometric margin is half the inter-interval distance. The one-sided form $\frac{1}{2}(\min_{X_+} f_{\text{dc}} - \max_{X_-} f_{\text{dc}})_+$ is the special case where X_+ lies above X_- in DC coordinate. Full proof in Appendix A. $\square$

The single-image case, where each class is one image together with all its shifts, is the observation of Ge et al. [10] that a shift-invariant linear classifier separates classes only through their DC component. What Theorem 1 adds is the exact convex-hull and interval form for an arbitrary finite separable dataset without orbit-closure, which is the form the architectural dissection of §5 measures, together with the extension to any finite symmetry group below.

The same accounting holds for any finite symmetry group, with the DC projector replaced by the group-average projector.

Theorem 2 (General finite-group invariant margin). *Let a finite group G act orthogonally on $\mathbb{R}^d$ by matrices $\{R_g\}$, with fixed subspace $V^G = \{v : R_g v = v \ \forall g\}$ and group-average projector $P_G = \frac{1}{|G|} \sum_{g \in G} R_g$, which is the orthogonal projector onto V^G . A linear score $w^\top x + b$ is exactly G -invariant if and only if $w \in V^G$, and the maximum invariant ℓ_2 margin for finite separable classes is*

$$\gamma_{\text{inv}} = \frac{1}{2} \text{dist}(\text{conv } P_G X_+, \text{conv } P_G X_-).$$

Theorem 1 is the cyclic case $P_G = \Pi_{\text{inv}}$. (Proof in Appendix A.)

3.2 The co-existence / trade-off dichotomy and the η/L criterion

Theorem 1 reduces “does invariance hurt robustness?” to “is the discriminative signal preserved by projection onto the invariant subspace?”. If the projected classes overlap, $\gamma_{\text{inv}} = 0$ (forced trade-off); if they stay separated by $\eta > 0$ with the invariant feature having Lipschitz constant L , the invariant classifier is simultaneously fully shift-consistent and robust with radius at least η/L . The governing scalar is therefore

$$\boxed{\eta/L = \text{Sep}_{\text{inv}}/L} \quad (\text{margin of the invariant feature, divided by its Lipschitz constant}).$$

We stress η/L , not merely $\text{Sep}_{\text{inv}} > 0$: invariance and robustness trade off whenever the margin that remains is small relative to its sensitivity, either because the projection discards the discriminative

signal or because it is preserved only at a vanishing margin. The Kamath et al. [18] construction is the first case (§3.4). The bound $r_2 \geq \eta/L$ is a Lipschitz–margin certificate: there is no *universal* converse, and the true robust radius can be arbitrarily larger than η/L (for the invariant feature $t \mapsto t^{2n+1}$ on the DC line $t = f_{dc}(x)$, $\eta/L = 1/(2n+1)$ while $r_2 = 1$, so the gap is unbounded). Under an explicit reachability condition, however, the certificate completes to a two-sided bracket.

Proposition 1 (Two-sided bracket: a conditional converse). *If the score is L -Lipschitz toward the boundary and admits a co-Lipschitz (lower-Lipschitz) constant α along a path reaching it, then $\eta/L \leq r_2 \leq \eta/\alpha$ with condition number $\kappa := L/\alpha \geq 1$, so η/L determines the robust radius up to κ . The bracket is exact ($r_2 = \eta/\|w\|_2$, $\kappa = 1$) for linear invariant features [14]; the t^{2n+1} example above is the large- κ regime, and for the power-spectrum surrogate both arms are explicit (verified numerically, realized $\kappa \approx 1.6$ –2.4). The certificate chain of Tsuzuku et al. [37] (their Eq. 5) bounds r_2 only from below, through successively tighter Lipschitz constants; the upper bound $r_2 \leq \eta/\alpha$ is the matching arm that turns the one-sided certificate into a two-sided bracket. A tight converse of a different kind is already known for randomized smoothing, where Cohen et al. [4] show the smoothing radius is the largest the class probabilities alone can certify and is exact for linear bases; our bracket is instead deterministic and tied to a single fixed classifier, with looseness controlled by the explicit condition number κ rather than by a worst case over classifiers. The full proof, with the explicit reaching-path and co-Lipschitz hypotheses, is in Appendix A.1.*

We therefore read η/L as a certificate and empirical predictor of robustness (§5) and, up to κ , as a characterization.

The excessive-invariance side. The ratio η/L lower-bounds model-score stability; a second quantity bounds whether the imposed invariance is semantically valid at the same radius. Let y_* be an oracle label and g a G -invariant classifier correct at x , and define the orbit-flip radius $\rho_G(x) = \inf\{\|R_h x - x\|_2 : y_*(R_h x) \neq y_*(x)\}$.

Proposition 2 (Orbit-flip upper bound). *If the set in the infimum is nonempty, the oracle-robust radius of g at x is at most $\rho_G(x)$: the orbit point $R_h x$ attaining it is correctly classified by g yet wrong under the oracle, an invariance-based adversarial example. (Proof in Appendix A.)*

A perturbation budget ϵ therefore admits a robust invariant classifier only if $\epsilon \leq \eta/L$ (score stability) and $\epsilon < \rho_G$ (the imposed invariance does not cross the oracle boundary). This is the formal core of the excessive-invariance failure of Tramèr et al. [34]: η/L controls sensitivity-based attacks, ρ_G controls invariance-based ones, and a model needs room under both.

3.3 Nonlinear escape: a convolution-plus-pooling layer computes the power spectrum

The invariant *linear* family is one-dimensional (only f_{dc}). Nonlinear invariants are richer.

Lemma 1 (Power-spectrum span). *A circular-convolution layer with a quadratic activation and global average pooling computes, for filter w , $\Phi_w(x) = \frac{1}{d} \sum_s ((w \star x)_s)^2 = \frac{1}{d} \sum_k |\hat{w}_k|^2 |\hat{x}_k|^2$. Varying a real filter w spans the nonredundant power coordinates $|\hat{x}_0|^2, |\hat{x}_{d/2}|^2$ (when d is even), and the conjugate-pair sums $|\hat{x}_k|^2 + |\hat{x}_{d-k}|^2$ for $0 < k < d/2$; since x is real these exhaust its power spectrum. (A complex filter isolates each individual $|\hat{x}_k|^2$.) These features are shift-invariant and supported on V_{ac} for $k \neq 0$.*

Proof. Cross-correlation has DFT $\widehat{w_k \star x_k}$; Parseval with the unitary DFT gives the displayed identity. A real filter satisfies $|\hat{w}_k|^2 = |\hat{w}_{d-k}|^2$, so it can weight a conjugate pair $\{k, d-k\}$ but not split it; concentrating it on one pair isolates $|\hat{x}_k|^2 + |\hat{x}_{d-k}|^2$, while a complex filter supported on a single frequency isolates $|\hat{x}_k|^2$. $\square$

Corollary 1 (Conditional robust radius). *For an invariant feature map Ψ with achievable head margin $\eta = \frac{1}{2} \text{dist}(\text{conv } \Psi(X_+), \text{conv } \Psi(X_-))$ and Lipschitz constant L , the induced classifier has SC = 1 and $r_2 \geq \eta/L$.*

This is the standard Lipschitz-margin bound; its content is which Ψ a shift-invariant architecture can realize (Lemma 1), not the inequality itself, and as noted in §3.2 it is a lower-bound certificate with no general converse. Here $\eta = \frac{1}{2} \text{dist}(\text{conv } \Psi(X_+), \text{conv } \Psi(X_-))$ is the exact hard-margin separation of the feature sets. On a ball $\|x\|_2 \leq B$ the power-spectrum feature is $2B$ -Lipschitz, which makes the certificate explicit, and Appendix A works out an exactly solvable two-subspace model whose

every point has robust radius $a/\sqrt{2}$ under a quadratic invariant that no linear classifier can match. A complementary first-order fact: for the continuous shift generator, an invariant score has gradient orthogonal to the orbit tangent, so first-order perturbations along the continuous orbit do not change the score; the discrete shift is a finite secant and is not protected exactly.

3.4 Recoveries and a corrected reconciliation

Ge et al. [10]. For the single white/black dot, $f_{dc}(\pm e_j) = \pm 1/\sqrt{d}$, so Theorem 1 gives $\gamma_{inv} = 1/\sqrt{d}$; in their one-sided convention the gap is $2/\sqrt{d}$ (their Corollary 1). The contrast “unconstrained margin 1 vs invariant $1/\sqrt{d}$ ” is the two-image protocol: it compares the unshifted pair $\{e_j\}, \{-e_j\}$ (unconstrained margin 1) against the invariant classifier. On the orbit-closed dataset $\mathcal{O}(e_j)$ vs $\mathcal{O}(-e_j)$ the unconstrained linear max-margin is itself $1/\sqrt{d}$, so the gap is a property of that protocol, not of invariance alone. The power spectrum is sign-blind ($|\widehat{+e_{j_k}}|^2 = |\widehat{-e_{j_k}}|^2$), so even the power-spectrum invariant cannot separate the dot, while the linear DC invariant separates it only at the weak margin $1/\sqrt{d}$: localized, phase-coded signal is the small- η/L corner for both. A higher-order invariant escapes it, however: the bispectrum $\hat{x}_{k_1}\hat{x}_{k_2}\hat{x}_{k_1+k_2}$ is also shift-invariant and separates $\pm e_j$ with margin $d^{-3/2}$ (it determines a generic signal up to shift), at the cost of a larger Lipschitz constant, hence a smaller certified η/L .

Kamath et al. [18]. Their construction embeds the robustness–accuracy distribution of Tsipras et al. [36] so that the discriminative signal is a binary cyclic code carried by the coordinates the shift permutes, each with ℓ_∞ margin of order $1/\sqrt{d}$; the single coordinate the shift fixes carries only weak signal (correlation $p \geq \frac{1}{2}$ with the label). A genuinely shift-invariant classifier can therefore use only that weak fixed coordinate and the power spectrum, and because the code is sign-coded its two classes have identical power spectra, so the invariant margin that remains is essentially zero. This is the same sign-blind, small- η/L corner as the Ge et al. [10] dot, now embedded in a statistical model: the trade-off appears precisely because they take $p \rightarrow \frac{1}{2}$, which empties the fixed coordinate. When p is bounded away from $\frac{1}{2}$, that coordinate is a strong invariant feature with $\Theta(1)$ margin and radius, which is our co-existence regime. So their impossibility is an instance of the η/L criterion rather than a counterexample. We state this as an interpretation: a self-contained restatement of their exact distribution would be needed to make it a theorem in our framework.

4 Optimization: does gradient descent reach the robust invariant solution?

The structural theory says robust shift-invariant classifiers exist; it does not say gradient descent finds them. For unconstrained two-layer ReLU networks on near-orthogonal cluster data, the implicit bias of gradient descent provably selects a non-robust max-margin solution even though robust ones exist [38, 7], via *feature averaging* [20]; Min and Vidal [25] conjecture that changing the architecture re-points this bias, and a natural hope is that hard-wired shift-invariance is such a lever. Testing that hope needs orbit-structured data on which the non-robustness premise genuinely holds. We characterize exactly when it can, via the rank of an orbit, and find that the two canonical orbit constructions both fail to, in opposite ways.

Lemma 2 (Orbit-bundle form). *A circular-convolution-plus-global-pooling network equals a two-layer network whose neurons are tied into shift-orbit bundles (one filter spawns all d of its shifts, sharing one output weight) and whose feature is an orbit average. (Exact; structural.)*

Lemma 3 (Rank of a shift orbit). *For a base pattern $u \in \mathbb{R}^d$, the cyclic orbit $\{S_s u\}_{s \in \mathbb{Z}_d}$ has rank exactly $\#\{k : \hat{u}_k \neq 0\}$, the number of nonzero Fourier coefficients of u .*

Proof. The matrix of orbit columns is circulant, hence diagonalized by the DFT with eigenvalues $\propto \hat{u}_k$; the rank is the number of nonzero eigenvalues. $\square$

Proposition 3 (The canonical orbit bases are degenerate, in opposite ways). *Let each class be the shift orbit of a single base pattern. (i) For a single cosine of frequency k , Lemma 3 gives orbit rank 2 (it spans $\{\cos, \sin\}$ at frequency k), violating the near-orthogonal $\Omega(d)$ -cluster assumption of Frei et al. [7], Li et al. [20]; moreover the signed orbit mean, the direction that carries the non-robustness conclusion in those works, is 0. (ii) For a localized spike e_j , the orbit is full-rank and orthonormal (the cluster assumption holds) but the power spectrum is phase-blind, so $\text{Sep}(\Psi) = 0$ and the invariant nonlinear classifier cannot separate the two classes. Thus neither canonical base is a valid testbed*

for the re-pointing hope: in (i) the non-robustness premise fails, in (ii) the invariant-separation conclusion fails.

Verification. For $d = 64$, $k = 5$: the cosine orbit has numerical rank 2 and signed mean of norm 1.7×10^{-16} ; the spike orbit has rank d and power-spectrum gap 0. $\square$

Proposition 4 (The degeneracy is non-generic). *If a base u has all Fourier coefficients nonzero, its orbit is full-rank (Lemma 3); and two such bases with distinct power spectra are power-spectrum separable. Hence generic single-orbit data is simultaneously full-rank and invariant-separable, so the degenerate corner of Proposition 3 does not cover single-orbit data in general.*

Lemma 3 and Propositions 3–4 are modest structural contributions: they pin down exactly which orbit data is degenerate and show the two textbook constructions are the degenerate cases (which explains why an unconstrained network is already competitive on them), while generic single-orbit data is not degenerate. They do *not* close the optimization gap, but a direct test on synthetic orbit data sharpens it. Training unconstrained versus weight-tied shift-invariant ReLU networks on the same orbit data shows the effect is *data-model-dependent*. On generic orbit data, shift-invariance is provably *margin-free*: on orbit-closed data the invariant and unconstrained two-layer-ReLU max-margin values coincide, because orbit-averaging an unconstrained solution gives an invariant one of equal minimum norm; so weight sharing cannot raise the margin there, and indeed the unconstrained network self-symmetrizes with no feature-averaging collapse. But on near-orthogonal cluster-geometry data the invariant architecture reaches a substantially larger η/L than the unconstrained one (a factor 1.2 in the rich regime, up to ~ 4 in the lazy regime, robust across 4–8 orbits and $d \in \{64, 128\}$), at equal margin: the gain is entirely a smaller Lipschitz constant, and it is largest exactly where the unconstrained network fails to self-symmetrize. So whether invariance re-points the implicit bias toward robustness depends on the data geometry. In the lazy/NTK regime the cluster-geometry case is now a theorem: orbit-averaging is the orthogonal projection onto the invariant subspace of the tangent RKHS [6], so the invariant min-norm interpolant has a smaller certified Lipschitz constant at equal margin, with the gap equal to the unconstrained solution’s orbit-variance [17] (Theorem 4, Appendix A.1). The rich (small-init) regime and the bare-ReLU trajectory remain open, on the same wall reached by Frei et al. [7], Min and Vidal [25], Li et al. [20]. Empirically (§5), η/L predicts the robustness of trained models regardless.

5 Experiments

Code, training configurations, trained checkpoints, and the per-cell results behind every correlation below are provided with the submission, so each reported number is reproducible end to end.

Structural predictions, training-free (confirmed). We compute, per dataset, the unconstrained linear margin $\text{Sep}(\text{Id})$, the invariant linear margin $\text{Sep}(\Pi_{\text{inv}})$ (Theorem 1), and the power-spectrum margin via a hard-margin SVM (Table 1). The single dot is separable by an unconstrained classifier (margin 1), only weakly by linear invariance ($1/\sqrt{d}$), and not at all by the power spectrum (sign-blind). The orthogonal-frequency data of Ge et al. [10] (each class the span of odd, resp. even, frequencies) is separable by *neither* linear classifier yet separable by the power-spectrum invariant: invariance helps through the right nonlinear feature.

Table 1: Achievable separation by classifier family (training-free), $d = 64$ so $1/\sqrt{d} = 0.125$.

dataset	Sep(Id)	Sep(Π_{inv})	η_{Ψ} (power spec.)	η_{Ψ}/L
dot ($+e_0 / -e_0$)	1.000	0.125	0.000	0
frequency, single	0.000	0.000	361.9	3.10
frequency, spans (Ge)	0.000	0.000	5.75	0.378

Architectural dissection (η/L predicts robustness; shift-consistency does not). We train seven architectures (a fully connected baseline; a single convolution-plus-global-pooling shift-invariant model; multi-layer convolutions with max/average pooling and an optional dense layer), each with and without batch-normalization where applicable, on 16,000 images for 25 epochs, three seeds, on GPU, for both MNIST and Fashion-MNIST. As the dataset-comparable robustness measure

we use the per-sample *robust radius* r_2 (the smallest ℓ_2 -PGD perturbation that flips the label), of which η/L is a lower bound; fixed- ϵ robust accuracy is misleading across datasets because of floor effects. Table 2 reports the MNIST per-model means. Across the seven configurations, η/L predicts the robust radius with Spearman $\rho = +0.96$ on MNIST and $+0.71$ on Fashion-MNIST, whereas shift-consistency predicts it on neither ($\rho = -0.31$ and 0.00). The single shift-invariant model has consistency 1.0 yet the smallest robust radius, the sharpest illustration that consistency does not imply robustness. Batch-normalization raises clean accuracy and lowers fixed- ϵ robust accuracy in both the fully connected and convolutional pairs, consistent with Galloway et al. [8]; under the per-sample robust radius, however, the change reverses sign and is small, which is precisely why we treat the radius, not fixed- ϵ accuracy, as the dataset-comparable measure.

Table 2: MNIST dissection (means over 3 seeds, $n=16,000$, 25 epochs). Primary metric: per-sample ℓ_2 robust radius r_2 (smallest label-flipping ℓ_2 -PGD perturbation), of which η/L is a lower bound. Fixed- ϵ robust accuracy (rob_ϵ , ℓ_2 at $\epsilon=1.5$) is shown for comparison.

model	params	clean	consist.	r_2	$\text{rob}_\epsilon \ell_2$	η/L
FC (depth 2)	269k	0.971	0.281	0.938	0.060	0.592
FC (depth 2) + BN	270k	0.973	0.292	0.962	0.043	0.668
conv d1 k5 GAP	4.6k	0.569	1.000	0.457	0.047	0.376
conv d1 k5 GAP + BN	4.9k	0.647	1.000	0.486	0.022	0.557
conv d2 k3 max	212k	0.987	0.473	1.846	0.477	1.443
conv d2 k3 avg	212k	0.984	0.475	1.658	0.399	1.537
conv d2 k3 max + dense	953k	0.986	0.487	1.944	0.570	1.627
Spearman over 7 rows: η/L vs $r_2 = +0.96$ (MNIST), $+0.71$ (Fashion); consist. vs $r_2 = -0.31, 0.00$.						

Capacity-matched CIFAR-10 dissection. We repeat the test on CIFAR-10 with a controlled comparison along the shift-invariance axis at matched capacity. Four architectures share one channel schedule, hence an identical parameter count, and differ only in the shift-invariance operator: a *standard* network (max-pooling, zero padding); an *anti-aliased* network with blur-pooled downsampling [40]; an *exact cyclic* network (circular padding, global pooling, no spatial subsampling) that is exactly invariant to integer circular shifts; and a *shift-augmented* network, the standard architecture trained with random circular shifts so its invariance is learned rather than built in. We train three seeds at two widths and report clean accuracy, circular-shift consistency, the per-sample ℓ_2 robust radius r_2 from the decoupled direction and norm (DDN) minimum-norm attack [30] (validated against the closed-form radius of a linear model to under one percent), and the active logit margin M (the gap between the correct-class logit and the highest other logit), summarized by its mean $\eta = \mathbb{E}[M]$ and mean input-gradient norm $L = \mathbb{E}[\|\nabla M\|_2]$ (Table 3). The threat-matched ratio $\eta/\|\nabla M\|_q$ used under adversarial training takes q the dual norm of the attack. Clean accuracy is comparable across arms, with the exactly invariant model slightly lower. AutoAttack [5] at the standard CIFAR thresholds drives every model to zero robust accuracy, as expected for models that are not adversarially trained; this floor is the reason we use the per-sample radius as the comparable quantity. At smaller ℓ_2 radii where AutoAttack is informative, the radius and AutoAttack agree: across arms the fraction of points with $r_2 > \epsilon$ matches AutoAttack robust accuracy at $\epsilon \in \{0.05, 0.10, 0.15, 0.25\}$ to a mean absolute gap of 0.014 (maximum 0.025), so the radius is not inflated by gradient masking.

The CIFAR-10 dissection makes two points. First, η/L predicts the robust radius across all eight capacity-matched cells (Pearson 0.998), while shift-consistency does not: the exactly invariant model has perfect consistency yet the smallest radius, the clearest separation of the two. Second, the decomposition shows the two robustness-changing operators act through different terms (Figure 3). Anti-aliasing leaves the margin essentially unchanged and lowers the Lipschitz constant ($\Delta \log L \approx -0.2$), gaining robustness through reduced input sensitivity. Exact cyclic invariance instead shrinks the margin ($\Delta \log \eta \approx -0.3$ to -0.4) and loses robustness, the real-data form of the Ge et al. [10] collapse: forcing invariance costs discriminative margin even while it makes the model perfectly shift-consistent. This is the structural criterion of §3.2 made operational: invariance helps robustness only when it does not cost margin.

Under adversarial training. To test whether the picture holds under a strong standardized attack, we PGD-adversarially train [21] the same four arms (ℓ_∞ , $\epsilon = 8/255$, 7 steps; two widths, two seeds) and evaluate with AutoAttack at the RobustBench thresholds (Table 4). Adversarial training makes

Table 3: Capacity-matched CIFAR-10 dissection (means over 3 seeds). All four arms have identical parameter counts at each width (288.7k at $w=32$, 1.15M at $w=64$). r_2 is the per-sample ℓ_2 robust radius (DDN, on correctly classified points); η and L are the mean active logit margin and its mean input-gradient norm. AutoAttack at $\ell_\infty=8/255$ and $\ell_2=0.5$ is 0.000 for every arm (the floor that motivates r_2).

arm	w	clean	consist.	r_2	η	L	η/L
standard	32	0.888	0.854	0.099	7.85	81.9	0.096
anti-aliased	32	0.900	0.887	0.116	7.75	66.4	0.117
exact cyclic	32	0.871	1.000	0.080	5.06	64.5	0.079
shift-aug	32	0.903	0.928	0.106	6.99	67.2	0.104
standard	64	0.912	0.891	0.119	8.34	71.3	0.117
anti-aliased	64	0.919	0.909	0.141	8.57	59.4	0.144
exact cyclic	64	0.897	1.000	0.090	6.42	73.0	0.088
shift-aug	64	0.921	0.943	0.117	8.75	75.1	0.117

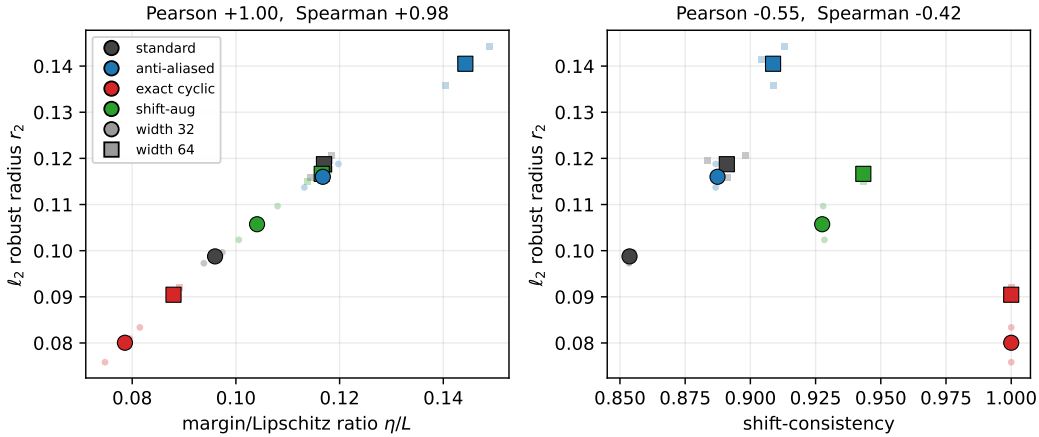


Figure 2: CIFAR-10, eight capacity-matched architecture-by-width cells (bold markers) and their three seeds (faint). Left: η/L tracks the ℓ_2 robust radius (Pearson 0.998, Spearman 0.98; 0.997 across individual seeds). Right: shift-consistency does not predict it; the exactly invariant model (red) has consistency 1.0 and the smallest radius.

AutoAttack informative: robust accuracy is now 26–42% at $\ell_\infty = 8/255$, with no gradient masking (AutoAttack stays below matched PGD for every arm). Three things hold. (i) Shift-consistency remains strongly *anti*-correlated with robustness (Pearson -0.88 against AutoAttack); the exactly invariant model has consistency 1.0 and the lowest robust accuracy of the four. (ii) The η/L criterion still predicts robustness once its norm is matched to the threat: the ℓ_∞ -matched ratio $\eta/\|\nabla M\|_1$, using the dual norm of the ℓ_∞ attack, tracks AutoAttack robust accuracy with Pearson 0.88 (Figure 4), whereas the ℓ_2 ratio (0.55) and the ℓ_2 robust radius (near zero) do not, because adversarial training increases robustness in ℓ_∞ , not ℓ_2 . (iii) The robustness ordering is anti-aliased $\geq$ standard $\approx$ shift-augmented $>$ exact cyclic, the same direction as standard training. One confound deserves attention: the exactly invariant arm reaches lower clean accuracy under adversarial training (0.57–0.63 vs. 0.69–0.78), so with only four arms one could read its robustness deficit as reduced fitting rather than margin geometry. The ResNet-scale dissection below settles this directly. It adds two parameter-matched low-invariance arms (zero-padding and an aliased max-pool variant) that sit at *low* consistency and *high* robustness, so the anti-prediction is carried by six arms spanning the full consistency range, not by the single exactly invariant outlier, and it holds when controlling for clean accuracy (partial correlation -0.82). The threat-matched η/L law is likewise not a clean-accuracy artifact.

Generality across datasets and threats. The two laws hold when we repeat the adversarial-training dissection on MNIST and Fashion-MNIST (ℓ_∞) and on CIFAR-10 under an ℓ_2 threat (Table 5; the per-arm robust accuracies behind these correlations are in Appendix A.2). Across all four conditions the threat-matched ratio $\eta/\|\nabla M\|_q$ (with q the dual norm of the attack) predicts AutoAttack robust

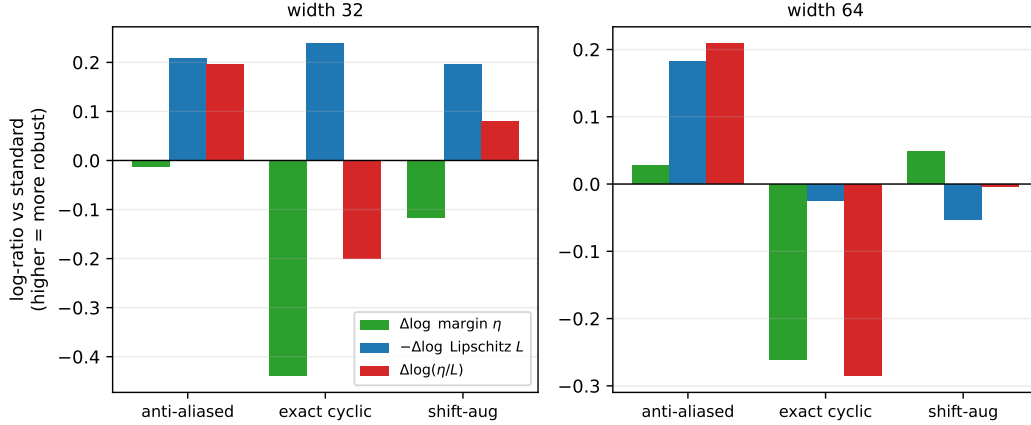


Figure 3: Margin/Lipschitz decomposition of the robust-radius change relative to the standard baseline, at each width ($\Delta \log(\eta/L) = \Delta \log \eta - \Delta \log L$; higher is more robust). Anti-aliasing raises η/L by lowering the Lipschitz constant with the margin roughly unchanged; exact cyclic invariance lowers η/L by shrinking the margin.

Table 4: PGD-adversarially trained CIFAR-10 dissection (means over 2 seeds; ℓ_∞ training at $\epsilon = 8/255$). AA_∞ and AA_2 are AutoAttack robust accuracy at $\ell_\infty=8/255$ and $\ell_2=0.5$; $\eta/\|\nabla M\|_1$ is the ℓ_∞ -matched margin-to-sensitivity ratio. Parameter counts per width are identical to Table 3.

arm	w	clean	consist.	AA_∞	AA_2	$\eta/\ \nabla M\ _1$	η/L_2
standard	32	0.712	0.794	0.375	0.531	0.0420	1.190
anti-aliased	32	0.724	0.802	0.388	0.540	0.0429	1.191
exact cyclic	32	0.569	1.000	0.261	0.402	0.0371	1.104
shift-aug	32	0.694	0.867	0.359	0.503	0.0417	1.149
standard	64	0.766	0.820	0.408	0.567	0.0411	1.133
anti-aliased	64	0.775	0.823	0.419	0.577	0.0423	1.154
exact cyclic	64	0.633	1.000	0.294	0.449	0.0365	1.080
shift-aug	64	0.750	0.891	0.395	0.548	0.0409	1.100

accuracy (up to +0.92), exceeding the mismatched ratio in every ℓ_∞ condition, and shift-consistency *anti*-predicts it (Pearson -0.44 to -0.92). The anti-prediction is weakest on MNIST: at $\ell_\infty=0.3$ every arm is roughly 94% robust, so the AutoAttack spread across arms is narrow and the consistency correlation, while still negative, is not significant at this grid size; the threat-matched η/L stays a strong predictor there (+0.92) because it does not saturate. The one place the matched ratio does not lead is ℓ_2 -CIFAR, where the two are comparable and the mismatched ratio is in fact marginally higher (matched +0.88, mismatched +0.91) because the ℓ_1 and ℓ_2 gradient norms run nearly proportional there; the threat-matching is then better read at the metric level, where the ℓ_2 robust radius predicts ℓ_2 AutoAttack (+0.80) even though it had decoupled under ℓ_∞ training. No gradient masking appears in any condition (AutoAttack $\leq$ matched PGD throughout), and the exact-cyclic arm has the lowest robust accuracy in every condition, again through a margin deficit ($\Delta \log \eta < 0$ for circular on all datasets).

Table 5: Adversarial-training dissection across datasets and threat models (four arms $\times$ two widths $\times$ two seeds; Pearson computed over the sixteen per-seed points against AutoAttack robust accuracy at the threat threshold). Matched = dual norm of the training threat (ℓ_1 gradient for ℓ_∞ , ℓ_2 gradient for ℓ_2). The consistency anti-prediction and threat-matched η/L prediction hold throughout.

dataset	threat	consist. vs AA	matched $\eta/\ \nabla M\ _q$	mismatched
CIFAR-10	ℓ_∞ 8/255	-0.88	+0.88	+0.55
CIFAR-10	ℓ_2 0.5	-0.81	+0.88	+0.91
MNIST	ℓ_∞ 0.3	-0.44	+0.92	+0.90
Fashion-MNIST	ℓ_∞ 0.1	-0.92	+0.71	+0.68

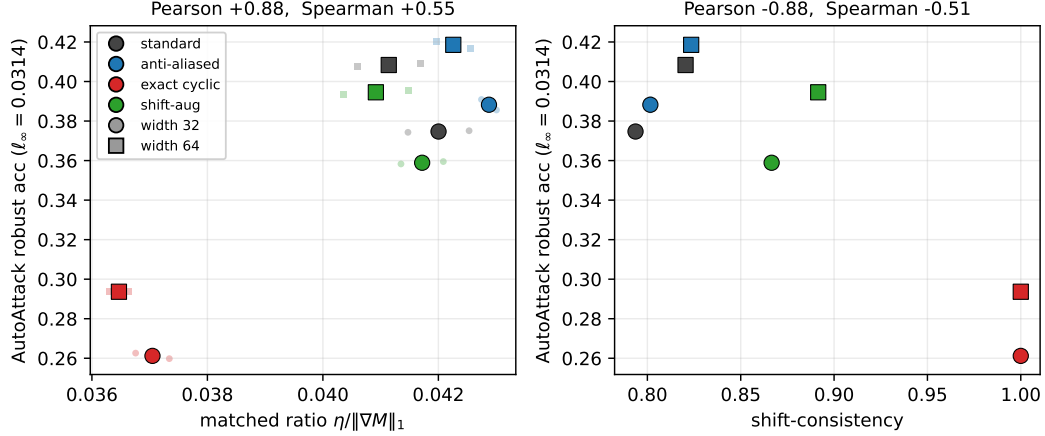


Figure 4: PGD-adversarially trained CIFAR-10 (eight cells, bold; two seeds, faint). Left: the ℓ_∞ -matched ratio $\eta/\|\nabla M\|_1$ tracks AutoAttack robust accuracy at $\ell_\infty = 8/255$ (Pearson 0.88). Right: shift-consistency anti-predicts it (Pearson -0.88); the exactly invariant model (red) has consistency 1.0 and the lowest robust accuracy.

Statistical power. Because each condition above has eight cells, we repeat the CIFAR-10 dissection on denser grids and report bootstrap 95% confidence intervals (resampling cells) with a two-sided permutation test. On a 24-cell standard-training grid (six widths, four seeds), shift-consistency is not a significant predictor of the robust radius (Pearson -0.33 , CI $[-0.68, +0.09]$, $p = 0.11$), consistent with the near-zero MNIST and Fashion-MNIST values, whereas η/L predicts it almost exactly ($+0.998$, CI $[+0.997, +0.999]$, $p < 10^{-3}$). The η/L -radius value is high by construction: η/L is a first-order lower bound on r_2 and these models are close to locally linear, so this correlation is the across-cell face of the per-sample certificate (§5) rather than a separate finding. The load-bearing result is the dissociation, that the architectural axis summarized by shift-consistency carries no information about the radius while the margin-to-sensitivity axis carries almost all of it. On a 12-cell adversarially trained grid (three widths, three seeds) the anti-prediction is significant: shift-consistency versus AutoAttack is -0.75 (CI $[-0.92, -0.02]$, $p = 0.006$) and the threat-matched ratio is $+0.83$ (CI $[+0.49, +0.93]$, $p = 0.002$), with the mismatched ratio not significant ($+0.27$, $p = 0.39$) and no gradient masking. The strong consistency anti-prediction is thus specific to adversarial training.

Scaling to a ResNet backbone. To check that the dissection is not an artifact of small networks, we repeat it on a PreActResNet-18 backbone on CIFAR-10 and CIFAR-100. We keep the four capacity-matched arms and add two parameter-matched low-invariance arms (a zero-padding variant and an explicitly aliased max-pool variant) at two widths and up to three seeds; here the exactly invariant arm uses adaptive polyphase sampling [2] rather than circular padding, so the exact-invariance result is reproduced through a second mechanism. All arms are PGD-adversarially trained at $\ell_\infty = 8/255$ with best-robust early stopping [29] and evaluated with AutoAttack (Figure 5). Both laws carry over at scale. Over the twelve architecture-by-width cells the threat-matched ratio $\eta/\|\nabla M\|_1$ predicts AutoAttack robust accuracy with Pearson $+0.91$ ($p < 10^{-3}$; $+0.77$ over the twenty-four individual trained models with a cell-resampling bootstrap), and under standard training η/L predicts the ℓ_2 robust radius with Pearson $+0.96$; shift-consistency anti-predicts robust accuracy with Pearson -0.81 ($p = 0.002$), and this persists when we control for clean accuracy (partial correlation -0.82). Across the four capacity-matched arms alone the anti-prediction is weak and not significant (-0.43); it sharpens to -0.81 only once the two low-invariance arms widen the consistency range. This is the point rather than a caveat: shift-consistency relates to robustness only across a broad enough span of the invariance axis and, even then, more weakly and with the opposite sign to what the anti-aliasing reading would suggest, whereas the threat-matched η/L predicts robustness throughout. No gradient masking appears (AutoAttack stays below matched PGD for every cell, including the non-differentiable adaptive-polyphase arm). On CIFAR-100 the threat-matched law holds in the same direction (Spearman $+1.0$ over the four arms), underpowered at four cells.

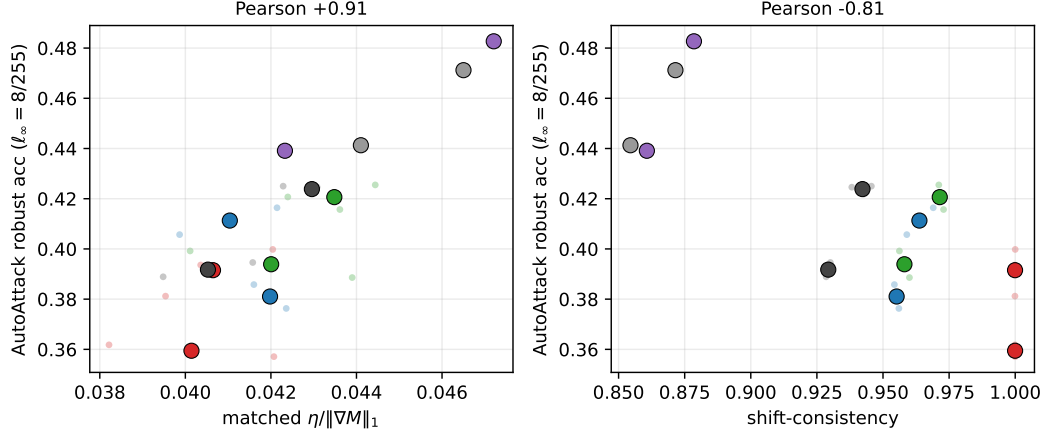


Figure 5: PGD-adversarially trained PreActResNet-18 on CIFAR-10 (twelve architecture-by-width cells, bold; seeds, faint). Left: the ℓ_∞ -matched ratio $\eta/\|\nabla M\|_1$ tracks AutoAttack robust accuracy at $\ell_\infty=8/255$ (Pearson +0.91). Right: shift-consistency anti-predicts it (Pearson -0.81); the exactly invariant adaptive-polyphase arm (red) has consistency 1.0 and the lowest robust accuracy, while the low-invariance arms (gray, purple) sit at low consistency and high robustness.

A mechanism for why shift-consistency is the unreliable signal is visible in the consistency axis. A ResNet backbone is already strongly shift-invariant, so over the four capacity-matched arms alone shift-consistency is squeezed into $[0.93, 1.00]$ and its anti-prediction washes out (Pearson -0.43 over eight cells, not significant); only when the low-invariance arms widen the axis to $[0.86, 1.00]$ does the anti-prediction become significant (-0.81). The margin-to-Lipschitz ratio does not saturate this way and separates the arms throughout: on a strong backbone shift-consistency is nearly constant while η/L still tracks robustness.

A graded anti-aliasing control at matched clean accuracy. The arms compared so far also differ in how well they fit, so we add a controlled sweep that varies anti-aliasing strength while holding clean accuracy fixed. We PGD-adversarially train the same backbone with blur-pool filters of increasing width (Rect-2, Tri-3, Bin-5, Bin-7), which raise shift-invariance at nearly constant clean accuracy, together with the standard (aliased) and exact-cyclic endpoints (six arms, two widths, two seeds; Figure 6). Three things follow. First, across the anti-aliasing family clean accuracy is matched within each width-and-seed cell to about one point (range 0.012), yet robust accuracy rises ($0.389 \rightarrow 0.407$): every anti-aliasing arm beats the aliased baseline in every cell (paired gain +0.014, bootstrap 95% CI $[+0.003, +0.027]$; the strongest, Bin-5, beats standard by +0.019, paired t -test $p = 0.007$), and the gain tracks a paired margin gain (η : +0.087, positive in every cell) at a roughly constant Lipschitz term. At matched fitting, anti-aliasing buys robustness through the margin, the opposite term to the one it acts on under standard training (where it instead lowers the Lipschitz constant, Figure 3). So invariance can help, and here it does so with no clean-accuracy cost. Second, shift-consistency is non-monotonic in robustness: it is nearly flat (≈ 0.80) across the anti-aliasing family and then jumps to 1.0 at the exact-cyclic arm, which is the *least* robust of the six, so higher consistency does not imply higher robust accuracy, whereas the threat-matched η/L orders the arms by robustness (Pearson +0.93). Third, and we state this plainly, the exact-cyclic arm’s robustness deficit cannot be separated from a clean-accuracy deficit: forcing exact invariance collapses the margin (η : $1.58 \rightarrow 0.86$), and that one collapse lowers clean accuracy ($0.74 \rightarrow 0.59$) and robust accuracy together, so across the six arms clean accuracy predicts robustness as strongly as η/L does. The right reading is therefore not that clean accuracy is a confounder to be removed, but that the margin is the upstream quantity both depend on: η/L tracks robustness because it is built on the margin the invariance operator changes, while shift-consistency, which ignores the margin, does not.

The certificate holds per sample, not only across cells. The correlations above are between cell means. The same ratio holds within a single model, sample by sample: read per input, the first-order ratio $M(x)/\|\nabla M(x)\|_2$ predicts that input’s own minimum-norm (DDN) robust radius with Spearman $\rho \approx 0.91$ under adversarial training and ≈ 0.95 under standard training, positive

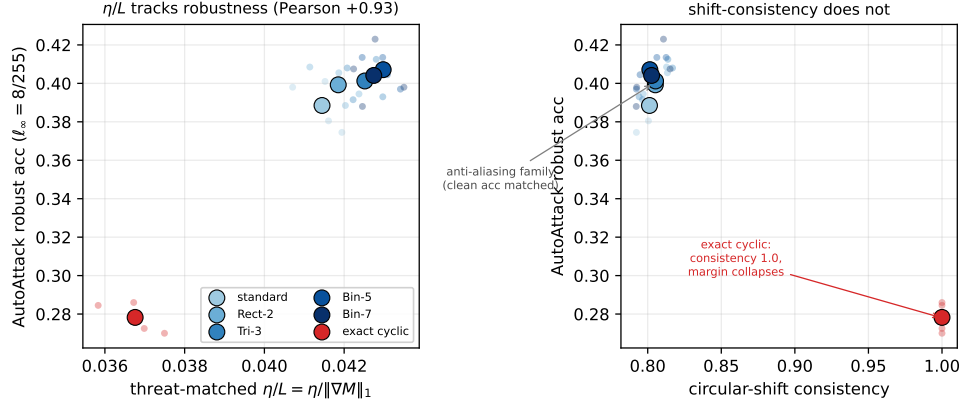


Figure 6: Graded anti-aliasing under PGD adversarial training (six capacity-matched arms; cell means bold, seeds faint). The anti-aliasing family (Rect-2 to Bin-7, blue) is matched on clean accuracy to within one point. Left: the threat-matched η/L orders all six arms by AutoAttack robust accuracy (Pearson +0.93). Right: shift-consistency does not – it is nearly flat across the anti-aliasing family and is 1.0 for the exact-cyclic arm (red), which is the least robust, because exact invariance collapses the margin. Clean accuracy co-varies with robustness across the arms through that same margin collapse.

in every one of the thirty-six trained CIFAR-10 models (twelve standard, twenty-four adversarial) and ≈ 0.97 on CIFAR-100 (Figure 7, left). The standard-training models are tighter because their smaller radii sit deeper in the locally linear regime where the first-order estimate is exact. This is the Lipschitz–margin certificate $r_2 \gtrsim M/\|\nabla M\|_2$ [14, 37] read pointwise; the first-order ratio is the small-curvature limit of the curvature-corrected bound of Moosavi-Dezfooli et al. [26], and the input-gradient norm in its denominator is the dimension-dependent vulnerability of Simon-Gabriel et al. [32]. Its strong rank agreement with the exact minimum-norm attack quantifies how locally linear the models are and shows η/L is a faithful per-sample robustness proxy, which is why the cell-mean correlation is so high.

Adversarial training inverts the within-model margin–sensitivity coupling. A second per-sample observation concerns how a model trades margin against input sensitivity across its own inputs. Under standard training the within-model rank correlation between a sample’s logit margin $M(x)$ and its input-gradient norm $\|\nabla M(x)\|_2$ is negative (Spearman ≈ -0.5 , all twelve standard-training models): high-margin points lie in flatter regions, so margin and robustness reinforce. Adversarial training flips the sign to positive ($\approx +0.5$, all twenty-four adversarially trained models; weaker on CIFAR-100), holding for the ℓ_1 gradient norm as well and persisting when we restrict to the above-median-margin half (Figure 7, right). We are not the first to relate logit margin to input sensitivity: that link is the Lipschitz–margin certificate, a margin consistency between logit and input-space margins has been documented in robust models [27], and adversarial training is already known to reshape the input–loss geometry by reducing curvature [26] and by aligning input gradients with semantic features [36, 16]. What is specific here is the *sign* of the per-sample coupling and its inversion: after adversarial training a model buys extra margin partly by raising local sensitivity, so the margin-to-sensitivity ratio is self-limiting, consistent with the modest robustness gains of forcing margin alone.

6 Related work

The two regimes. Whether shift-invariance helps or harms adversarial robustness has been argued in both directions. On the harming side, Ge et al. [10] prove that a fully shift-invariant classifier can be far less robust than a fully connected one. On the helping side, reducing aliasing has been shown to improve adversarial robustness in practice [12, 13, 28], and anti-aliased or learned-invariant pooling improves shift-consistency together with clean accuracy and corruption robustness [40, 31, 24], and most recently Wang et al. [39] build rotation- and scale-equivariant defenses and argue, through an

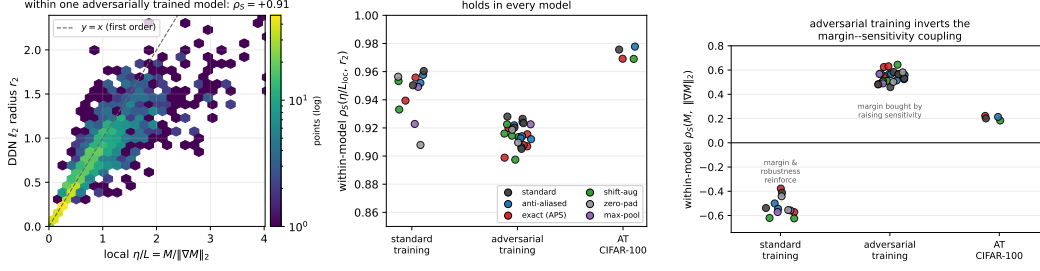


Figure 7: Per-sample structure on the PreActResNet-18 models. Left: within one adversarially trained model, the first-order ratio $M/\|\nabla M\|_2$ predicts each point’s own ℓ_2 robust radius (Spearman $+0.91$); the right inset plots that within-model rank correlation for each model and it is positive in every one, across standard training, adversarial training, and CIFAR-100. Right: the within-model coupling between margin M and input-gradient norm $\|\nabla M\|_2$ is negative under standard training and positive after adversarial training, for every model.

orbit-invariant Lipschitz bound, that symmetry improves certified robustness. These findings have not been placed on a common axis; we do so through whether the discriminative signal is preserved under projection onto the invariant subspace with positive margin, summarized by η/L . A related trade-off line shows that geometric (spatial) invariance and adversarial robustness cannot both be maximized [18, 33], and that more invariance raises vulnerability to invariance-based attacks [34, 9]; in our terms these are the small- η/L and large- ρ_G corners of the same criterion (§3.2).

What we add. The Lipschitz–margin certificate $r_2 \geq \eta/L$ is classical [14, 37], with deterministic [19] and randomized-smoothing [4] descendants, and the per-sample link between logit margin and input-space margin has been documented in robust models [27]. Our contribution is not the bound but the shift-invariance dissection it supports: that shift-consistency does not predict robustness while a threat-matched η/L does, and that adversarial training inverts the within-model margin–sensitivity coupling. The orbit-invariant Lipschitz mechanism of Wang et al. [39] is, in this taxonomy, the shift-consistency ingredient we find necessary but not sufficient: our lazy-regime theorem (Appendix A.1) shares their observation that invariance can lower the certified Lipschitz constant, but adds the exact orbit-variance gap, the matching upper bound on the radius, and the margin-side account of the regime where invariance instead destroys the margin and hurts. On the optimization side, gradient descent provably converges to non-robust networks [38, 7, 20], Min and Vidal [25] conjecture an architectural fix, and Melamed et al. [23] tie non-robustness to off-manifold gradients; the neural-tangent-kernel account of robust versus non-robust features [35] underlies our lazy-regime result, and invariance has separately been used to aid robustness directly, for instance through test-time equivariance [22]. Architecture-versus-robustness studies [15] vary kernel, normalization, depth, and width under strong attacks but not shift-consistency. The robustness background is Madry et al. [21], Goodfellow et al. [11], Tsipras et al. [36].

7 Limitations and open problems

The central open problem is a trajectory-level implicit-bias theorem for shift-invariant convolution-plus-pooling networks on data where the non-robustness premise genuinely holds (Proposition 3 shows the canonical orbit constructions are degenerate, while Proposition 4 shows generic orbit data is not, so such data exists). The criterion η/L is a lower-bound certificate with no general converse, so the true robust radius can exceed it; the power-spectrum Lipschitz constant grows with input norm, so the guarantee is normalization-dependent; for real filters the convolution-square-pool layer spans conjugate-pair power coordinates rather than individual frequencies; the orbit-gradient statement is exact only for the continuous group; and the empirical study covers standard and PGD-adversarial training across three datasets, two threat models, and small-convolutional and PreActResNet-18 backbones, so still larger architectures and the rich-regime trajectory analysis are left to future work.

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A Proofs and additional results

Proof of Theorem 1. Invariance of $g(x) = w^\top x + b$ under every shift means $w^\top S_s x = w^\top x$ for all x, s , i.e. $(S_{-s}w - w)^\top x = 0$ for all x , so $S_{-s}w = w$ for all s ; the only vectors fixed by every cyclic shift are the constants, hence $w \in V_{\text{inv}} = \text{span}(\mathbf{1})$. Writing $w = c\mathbf{1}$, the score depends on x only through $f_{\text{dc}}(x) = \mathbf{1}^\top x / \sqrt{d}$, so the data reduce to the scalar DC values and the classes to the intervals $I_\pm$. The maximum-margin separator of two point sets on a line places the threshold midway between the nearer interval endpoints, with geometric margin half the distance between the intervals, giving $\gamma_{\text{inv}} = \frac{1}{2} \text{dist}(I_+, I_-)$. Identifying the projected line with $\text{span}(\mathbf{1})$ gives the convex-hull form. $\square$

Proof of Theorem 2. P_G is the orthogonal projector onto V^G : by group closure $P_G^2 = |G|^{-2} \sum_{g,h} R_g R_h = |G|^{-1} \sum_k R_k = P_G$; each R_g is orthogonal so $R_g^\top = R_{g^{-1}}$ and $P_G^\top = P_G$; its range is exactly the fixed subspace V^G . Invariance of $w^\top x$ for all x is equivalent to $R_g^\top w = w$ for all g , hence (as G is a group) to $R_g w = w$, i.e. $w \in V^G$. For $w \in V^G$ we have $w = P_G w$, so $w^\top x = w^\top P_G x$: every invariant linear classifier on $X_\pm$ is an ordinary linear classifier on the projected sets $P_G X_\pm$, with the same normal and signed margins. The maximum unit-normal margin separating two finite sets A, B equals $\frac{1}{2} \text{dist}(\text{conv } A, \text{conv } B)$, attained by the unit vector along the closest-points segment of the hulls; applying this to $A = P_G X_+, B = P_G X_-$ gives the claim. The cyclic case has $P_G = \Pi_{\text{inv}}$. $\square$

Proof of Proposition 2. For any h with $y_*(R_h x) \neq y_*(x)$, invariance gives $\text{sign } g(R_h x) = \text{sign } g(x) = y_*(x) \neq y_*(R_h x)$. Thus $R_h x$ is classified by g as $y_*(x)$ but the oracle label there is different, so $R_h x$ is an oracle-adversarial example at distance $\|R_h x - x\|_2$. Taking the infimum over such h bounds the oracle-robust radius by $\rho_G(x)$. $\square$

Lemma 4 (Power-spectrum Lipschitz bound on a ball). *Let $Q(x) = (|\hat{x}_k|^2)_k$ with the unitary DFT. On $\|x\|_2 \leq B$, $\|Q(x) - Q(y)\|_2 \leq 2B \|x - y\|_2$, and a scalar quadratic invariant $f_a(x) = \sum_k a_k |\hat{x}_k|^2$ has $\|\nabla f_a(x)\|_2 \leq 2B \|a\|_\infty$.*

Proof. For complex z, z' , $||z|^2 - |z'|^2| \leq |z - z'|(|z| + |z'|)$. Applying this coordinatewise to Fx, Fy and using unitarity of F ,

$$\|Q(x) - Q(y)\|_2 \leq \|F(x - y)\|_2 (\|Fx\|_2 + \|Fy\|_2) = \|x - y\|_2 (\|x\|_2 + \|y\|_2) \leq 2B\|x - y\|_2.$$

For the gradient, $\nabla f_a(x) = 2F^* \text{diag}(a)Fx$ in the real identification, so $\|\nabla f_a(x)\|_2 \leq 2\|a\|_\infty \|x\|_2 \leq 2B\|a\|_\infty$. $\square$

Theorem 3 (An exactly solvable nonlinear invariant model). *Let U, V be orthogonal subspaces of $\mathbb{R}^d$ and let $X_+ = \{u \in U : \|u\|_2 = a\}$, $X_- = \{v \in V : \|v\|_2 = a\}$ (or finite shift-orbit subsets whose hulls contain 0). The quadratic invariant $f(x) = \|P_U x\|_2^2 - \|P_V x\|_2^2$ classifies X_+, X_- correctly, every point has exact ℓ_2 robust radius $a/\sqrt{2}$, and no linear classifier separates the classes.*

Proof. For $x \in X_+$, $P_V x = 0$ and $\|x\|_2 = a$, so $f(x) = a^2 > 0$. Decompose a perturbation as $\delta = \delta_U + \delta_V + \delta_W$ with $W = (U \oplus V)^\perp$. The boundary condition $f(x + \delta) \leq 0$ reads $\|x + \delta_U\|_2^2 \leq \|\delta_V\|_2^2$; δ_W cannot help, so set it to 0. With $r = \|x + \delta_U\|_2$, the smallest $\|\delta_U\|_2$ is $|a - r|$ (move along the ray through x) and the smallest feasible $\|\delta_V\|_2$ is r , so $\|\delta\|_2^2 \geq (a - r)^2 + r^2$, minimized at $r = a/2$ with value $a^2/2$; hence the radius is $a/\sqrt{2}$, attained. The case $x \in X_-$ is symmetric. No affine hyperplane separates the classes because 0 lies in both class convex hulls, so the hulls intersect. $\square$

A.1 The two-sided bracket and the lazy-regime cluster theorem

We give in full the two results stated in the body: the conditional converse that completes the Lipschitz–margin certificate to a two-sided bracket (Proposition 1, §3.2), and the lazy-regime cluster advantage that the optimization section invokes (§4).

Proof of Proposition 1. Write $y \in \{\pm 1\}$ for the label of x , the margin $\eta := yg(x) > 0$, the decision boundary $\mathcal{B} = \{z : g(z) = 0\}$, and the robust radius $r_2 := r_2(g, x) = \inf\{\|\delta\|_2 : \text{sign } g(x + \delta) \neq y\}$. By hypothesis g is continuous and L -Lipschitz in ℓ_2 on a convex set S that contains x , the point of $\mathcal{B}$ nearest x , and the path γ below; convexity of S keeps the relevant segments inside S , so the local Lipschitz and co-Lipschitz constants apply along them.

Lower arm $r_2 \geq \eta/L$. For any δ with $\|\delta\|_2 < \eta/L$, the Lipschitz bound gives $|g(x + \delta) - g(x)| \leq L\|\delta\|_2 < \eta = |g(x)|$, so $g(x + \delta)$ has the same sign as $g(x)$ and $\text{sign } g(x + \delta) = y$. No perturbation of norm below η/L flips the label, hence $r_2 \geq \eta/L$ (the certificate of Corollary 1; 14, 37).

Upper arm $r_2 \leq \eta/\alpha$. Let $\gamma : [0, T] \rightarrow S$ be a continuous path with $\gamma(0) = x$, $g(\gamma(T)) = 0$ (so T is the first time γ meets $\mathcal{B}$), and a co-Lipschitz (lower-Lipschitz) bound $|g(\gamma(t)) - g(x)| \geq \alpha\|\gamma(t) - x\|_2$ holding up to $t = T$. Evaluating at $t = T$, where $g(\gamma(T)) = 0$ and $|g(x)| = \eta$,

$$\eta = |g(x) - g(\gamma(T))| \geq \alpha\|\gamma(T) - x\|_2, \quad \text{so} \quad \|\gamma(T) - x\|_2 \leq \eta/\alpha.$$

At $\gamma(T) \in \mathcal{B}$ we have $g(\gamma(T)) = 0$, so $\text{sign } g(\gamma(T)) \neq y$, and by continuity of g points immediately beyond $\gamma(T)$ have strictly opposite sign; hence the perturbation $\delta = \gamma(T) - x$ is label-flipping and $r_2 \leq \|\gamma(T) - x\|_2 \leq \eta/\alpha$.

$\alpha \leq L$ and the condition number. Since $g(\gamma(T)) = 0 \neq g(x)$ we have $\gamma(T) \neq x$, and the same secant obeys $\alpha\|\gamma(T) - x\|_2 \leq |g(\gamma(T)) - g(x)| \leq L\|\gamma(T) - x\|_2$, so $\alpha \leq L$ and $\kappa := L/\alpha \geq 1$. Combining the two arms, $\eta/L \leq r_2 \leq \eta/\alpha$, with $r_2 = \eta/L$ exactly when $\kappa = 1$.

Linear (exact) case. If $g(x) = w^\top x + b$ is linear, in particular an invariant linear feature ($w \in V_{\text{inv}}$, so g reads x only through f_{dc}), then for every z , $|g(z) - g(x)| = |w^\top(z - x)| \leq \|w\|_2 \|z - x\|_2$, with equality along the straight path $\gamma(t) = x - tw/\|w\|_2$. Thus $L = \alpha = \|w\|_2$, $\kappa = 1$, and the boundary is reached at distance $\eta/\|w\|_2$, giving the exact point-to-hyperplane radius $r_2 = \eta/\|w\|_2$ [14]. The unbounded- κ end is the $t \mapsto t^{2n+1}$ feature of §3.2 ($\eta/L = 1/(2n+1)$ while $r_2 = 1$): no universal converse exists, but every model sits in the bracket at its own finite or infinite κ . The deterministic bracket complements the randomized-smoothing converse of Cohen et al. [4], whose looseness is a worst case over classifiers rather than the explicit per-classifier κ here. $\square$

Theorem 4 (Lazy-regime cluster advantage: invariance lowers the certified Lipschitz constant). *Work in the lazy/NTK regime, where each trained network equals the minimum-RKHS-norm interpolant of*

its fixed neural tangent kernel [17]: write $K(x, x') = \langle \Phi(x), \Phi(x') \rangle_{\mathcal{H}}$ for the tangent kernel with feature map Φ and RKHS $\mathcal{H}_K$, and assume a smooth activation so that Φ is L_Φ -Lipschitz on the data ball $\{\|x\|_2 \leq B\}$. Let the data be orbit-closed with labels constant on orbits, let the kernel be shift-invariant ($K(S_s x, S_s x') = K(x, x')$ for all $s \in \mathbb{Z}_d$), and let $\Pi_G f = \frac{1}{d} \sum_{s \in \mathbb{Z}_d} f(S_s \cdot)$ be the orbit-averaging (Reynolds) operator, whose range $\mathcal{H}^G = \Pi_G \mathcal{H}_K$ is the invariant subspace realized by the shift-tied orbit bundle of Lemma 2. For the unit-margin min-norm problem $\min_f \|f\|_K$ s.t. $y_i f(x_i) \geq 1$, let f_{unc} be the solution over $\mathcal{H}_K$ and f_{inv} the solution over $\mathcal{H}^G$, and set the certified Lipschitz constants $L_{\text{unc}} = L_\Phi \|f_{\text{unc}}\|_K$, $L_{\text{inv}} = L_\Phi \|f_{\text{inv}}\|_K$. Then

- (i) Π_G is the orthogonal projection of $\mathcal{H}_K$ onto $\mathcal{H}^G$, hence non-expansive, with $\|f\|_K^2 = \|\Pi_G f\|_K^2 + \|(I - \Pi_G)f\|_K^2$ [6];
- (ii) $\Pi_G f_{\text{unc}}$ is feasible for the invariant problem, so $\|f_{\text{inv}}\|_K \leq \|\Pi_G f_{\text{unc}}\|_K \leq \|f_{\text{unc}}\|_K$;
- (iii) therefore $L_{\text{inv}} \leq L_{\text{unc}}$, and at the common unit margin η the invariant model has the weakly larger certified ratio, $\eta/L_{\text{inv}} \geq \eta/L_{\text{unc}}$. The gap is the discarded non-invariant energy $\|(I - \Pi_G)f_{\text{unc}}\|_K^2$ (the orbit-variance of the unconstrained solution), and is strict unless f_{unc} is already shift-invariant.

Proof. (i) For a shift-invariant kernel, Π_G is self-adjoint and idempotent on $\mathcal{H}_K$ with spectrum in $\{0, 1\}$, hence the orthogonal projection onto its fixed-point set $\mathcal{H}^G$; the displayed identity is the Pythagorean decomposition for that projection [6]. (ii) Because the data is orbit-closed with orbit-constant labels, each $S_s x_i$ is a training point of label y_i , so $y_i f_{\text{unc}}(S_s x_i) \geq 1$ for all s , and averaging gives $y_i (\Pi_G f_{\text{unc}})(x_i) \geq 1$. Thus $\Pi_G f_{\text{unc}} \in \mathcal{H}^G$ is feasible for the invariant min-norm problem, whence $\|f_{\text{inv}}\|_K \leq \|\Pi_G f_{\text{unc}}\|_K$, which (i) bounds above by $\|f_{\text{unc}}\|_K$. (iii) For any $f \in \mathcal{H}_K$ and inputs x, x' in the data ball, $|f(x) - f(x')| = |\langle f, \Phi(x) - \Phi(x') \rangle_K| \leq \|f\|_K \|\Phi(x) - \Phi(x')\|_K \leq L_\Phi \|f\|_K \|x - x'\|_2$, so f is $L_\Phi \|f\|_K$ -Lipschitz there. Applying this to f_{inv} and f_{unc} and using (ii) gives $L_{\text{inv}} \leq L_{\text{unc}}$; dividing the common unit margin η by these constants yields $\eta/L_{\text{inv}} \geq \eta/L_{\text{unc}}$. The orthogonality identity of (i) makes the norm gap exactly $\|(I - \Pi_G)f_{\text{unc}}\|_K^2$, which vanishes iff $f_{\text{unc}} = \Pi_G f_{\text{unc}}$ is already invariant. $\square$

Remark 1 (Reading and smooth-activation caveat). The norm contraction (i)–(ii) is activation-independent, but step (iii) uses smoothness: the certified constant compared here is the RKHS-norm bound $L_\Phi \|f\|_K$ on the local Lipschitz constant, not the realized constant. For a bare two-layer ReLU the tangent feature map is not Lipschitz [1] (its RKHS contains unit-norm functions of unbounded Lipschitz constant), so the theorem requires a smooth activation, or an explicit $\sup_{\|x\|_2 \leq B} \|\nabla \Phi(x)\|$ bound carried in place of L_Φ . The result thus proves the cluster-geometry advantage in the lazy, smooth-activation regime, exactly where the synthetic stress test shows the largest tied-versus-unconstrained η/L gap, while the rich (small-init) and bare-ReLU trajectories, where the fixed-kernel picture breaks down [3], stay open.

A.2 Per-arm robust accuracies under adversarial training

Table 5 reports the correlations that summarize the generality experiment of §5; Table 6 gives the per-arm AutoAttack robust accuracies behind them, with clean accuracy and circular-shift consistency alongside. The numbers make the saturation on MNIST explicit: at $\ell_\infty = 0.3$ every arm is robust to within one point of 94%, so the across-arm spread that the consistency correlation reads is narrow, which is why that correlation, though still negative, is not significant at this grid size. On Fashion-MNIST and ℓ_2 -CIFAR the spread is several times wider and the exactly invariant arm, whose consistency is 1.0 by construction, is the least robust in each width.

Table 6: Per-arm AutoAttack robust accuracy under PGD adversarial training for the three generality conditions (means over two seeds). MNIST and Fashion-MNIST are ℓ_∞ -trained and evaluated at the listed threshold; CIFAR-10 is ℓ_2 -trained and evaluated at $\ell_2 = 0.5$. Clean accuracy and circular-shift consistency are shown for reference. The four arms share a channel schedule and so an identical parameter count at each width.

arm	w	clean	consist.	robust acc.
<i>MNIST</i> , $\ell_\infty = 0.3$				
standard	32	0.994	0.945	0.942
anti-aliased	32	0.994	0.953	0.942
exact cyclic	32	0.994	1.000	0.926
shift-aug	32	0.994	0.998	0.939
standard	64	0.994	0.946	0.947
anti-aliased	64	0.994	0.954	0.947
exact cyclic	64	0.994	1.000	0.934
shift-aug	64	0.995	0.999	0.952
<i>Fashion-MNIST</i> , $\ell_\infty = 0.1$				
standard	32	0.876	0.684	0.776
anti-aliased	32	0.876	0.683	0.776
exact cyclic	32	0.855	1.000	0.724
shift-aug	32	0.862	0.958	0.747
standard	64	0.886	0.680	0.781
anti-aliased	64	0.885	0.690	0.783
exact cyclic	64	0.868	1.000	0.736
shift-aug	64	0.870	0.964	0.760
<i>CIFAR-10</i> , $\ell_2 = 0.5$				
standard	32	0.804	0.829	0.581
anti-aliased	32	0.815	0.832	0.594
exact cyclic	32	0.711	1.000	0.464
shift-aug	32	0.794	0.902	0.570
standard	64	0.846	0.847	0.609
anti-aliased	64	0.854	0.857	0.628
exact cyclic	64	0.769	1.000	0.511
shift-aug	64	0.841	0.924	0.613